

DHAWAL DIXIT

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EDUCATION

The University of Chicago

MS in Applied Data Science

Chicago, IL

Exp. Dec 2026

Coursework

Machine Learning • Statistical Modeling • Probabilistic Graphical Models • Bayesian ML • Online & Reinforcement Learning • Algorithms • Time Series Forecasting • Planning & Estimation for Robotics • NLP • Linear Algebra

Manipal Academy of Higher Education

B.Tech Electrical & Electronics (Minor in Data Science)

Manipal, India

Jul 2019 – May 2023

PROFESSIONAL EXPERIENCE

Wafra

Summer Analyst

New York, NY

Jun 2026 – Aug 2026

- Consolidated six private equity data sources on **50+** General Partners into a Microsoft Fabric lakehouse and a Power BI dashboard that became the firm's primary GP-tracking tool for **50+ users**, saving an estimated **~10 hours/week** per investment professional.
- Extracted structured fund metrics from **~40,000** manager documents (**5+ TB**) at **~95%** accuracy, with Spark pipelines routing each document type to deterministic parsing or GPT-5-mini OCR-based extraction, then mastered fund and contact entities across **5M+ records**, resolving investment vehicles and naming variants with Levenshtein and Jaro-Winkler distances via RapidFuzz.
- Built LLM-as-judge evaluation pipelines** and validation loops to monitor extraction quality, and wrote **5 Claude skills** for self-serve document extraction, summarization and deck generation, used by the **10-person** investment team for ad hoc analysis outside the pipelines.

Hewlett-Packard Enterprise

Systems Software Engineer

Bangalore, India

Aug 2023 – Aug 2025

- Raised weekly support case automation from **66% to 82%** and cut **\$100K+** in annual licensing by replacing a PolyAnalyst and SSIS chain with a single LightGBM classifier over six case types, deployed as a scheduled Dataiku scenario running Spark on HDFS.
- Engineered **~50 features** from multilingual text embeddings and structured attributes, and benchmarked LightGBM against logistic regression, XGBoost and deep networks on per-class F1 over **ten years** of labeled cases, unlocking Japanese cases the old system missed.
- Drove a reported **73%** gain in case resolution efficiency and a **30%** cut in external tooling across **four support teams** with LLM pipelines summarizing months-long case histories and scoring sentiment, run on Kubeflow with Langfuse tracing and drift monitoring.
- Fine-tuned FLAN-T5** on GPT-4o-labeled summaries and benchmarked it with Granite 3.2 and Llama 3 against GPT-4o, which won on engineer-validated quality and latency; stripped PII with NER and regex before data left the cluster, and enforced JSON schemas.

Hewlett-Packard Enterprise

Software Engineering Intern

Bangalore, India

Jan 2023 – Jul 2023

- Operated the team's **26-node, 300+ TB** HDFS cluster, HPE's go-to source for financial, legal, sales, factory and support data, handling YARN queue management, permissions and resource allocation, storage scaling and cleanup, and MapReduce job failure triage.
- Automated **12+** data integrity checks between upstream data marts and the data lake, and cut statistical process control pipeline runtime by **80%** through a refactor that made anomalies in factory data feeds easier to surface.

Haptik AI

Data Engineering Intern

Mumbai, India

Jun 2022 – Oct 2022

- Cut data warehouse read latency by **50%** by redesigning the schema for **10+ TB** of conversational message data from a flat layout to a hybrid normalized model, fast enough that product teams presented dashboards on live data in CTO meetings.

TECHNICAL SKILLS

Languages | Python • SQL • R • JavaScript • C++ • C#

ML & LLMs | PyTorch • PyTorch Geometric • JAX/Flax/Optax • NumPyro • scikit-learn • LightGBM • XGBoost • TensorFlow
Azure OpenAI • FLAN-T5 fine-tuning • Transformers • spaCy • Langfuse • vLLM • RapidFuzz • Claude Code/Skills

Data & MLOps | Spark • Spark SQL • Delta Lake • Microsoft Fabric • Airflow • Hive/Impala • HDFS/YARN • SQL Server (T-SQL)
MLflow • Dataiku • Kubeflow • Kubernetes • Linux • Modal • Weights & Biases • Power BI

RESEARCH EXPERIENCE

MS Thesis: Model Selection for Hybrid Graph Neural Networks | PyTorch Geometric | code

Apr 2026 – Present

- Studying why graph neural networks struggle to pass information between distant nodes, and when adding a global virtual node or attention layer to local message passing actually fixes it, so practitioners can pick the right hybrid GNN architecture for the size and structure of their graph.
- On a **35K-node** Reddit hyperlink graph, a virtual-node hybrid cut error vs a GraphSAGE baseline by **51%** at 3 hops and **81%** at 5+ hops over **100** seeds; Jacobian and attention-entropy probes on LRGB peptides then confirmed that both over-squashing and attention dilution were at work.

SELECT PROJECTS

Stochastic Trajectory Prediction via Motion Indeterminacy Diffusion | PyTorch, Modal | code

Dec 2025

- Rebuilt a published diffusion model for predicting where pedestrians will walk, writing the transformer denoising network from scratch on a Trajectron++ encoder that conditions each person's path on neighbors within 3 m. Trained on Modal GPUs and matched the paper's accuracy on **all five** ETH/UCY benchmark scenes to within **4 cm**, while cutting sampling from 100 denoising steps to just **10** by switching to a DDIM sampler.

Bayesian Failure Regimes in Cluster Traces | NumPyro, JAX | code

Mar 2026

- Modeled how jobs on Google's Borg cluster drift through hidden health states before failing, using a Bayesian hidden Markov model in NumPyro/JAX fitted with variational inference, choosing the number of states by ELBO and WAIC. Without ever seeing failure labels, the model found one stalled, low-utilization state that held **81%** of all task kills, a candidate early-warning signal for cluster operators to act on.

Sparse Autoencoders as MAP Inference | PyTorch | code

May 2026

- Showed that sparse autoencoders, a core tool for interpreting LLM internals, are MAP inference under a Laplace prior, and replaced them with variational autoencoders using a closed-form Gaussian-Laplace KL. SAEs still reconstructed better, but VAEs detected known features far more reliably (AUROC **0.84** vs **0.58**) and spread their belief across features on ambiguous inputs instead of guessing; written up as a paper.